

11	(a) changing voltage changes energy / speed of <u>electrons</u> changing electron energy changes maximum X-ray photon energy	M1 A1	[2]
(b)	(i) 1. loss of power / energy / intensity	B1	[1]
	2. intensity changes when beam not parallel decreases when beam is divergent	C1 A1	[2]
	(ii) ratio = $(\exp \{-2.9 \times 2.5\}) / (\exp \{-0.95 \times 6.0\})$ = 0.21 (<i>min. 2 sig. fig.</i>)	C1 A1	[2]
	<i>(values of both lengths incorrect by factor of 10^{-2} to give ratio of 0.985 scores 1 mark)</i>		

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